Plotting X as a Function of Y

Introduction
Plotting X as a Function of Y Using a Table of Values
Plotting X as a Function of Y Using a Reflection
Plotting X as a Function of Y Given the Graph of the Corresponding Function
Plotting a Quadratic Function Where X Is a Function of Y

Prerequisites

Roots of Quadratic Functions
Graphs of Inverse Functions

Introduction

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We're used to plotting curves where y is a function of x. However, we can also plot curves where x is a function of y.

For example, consider the curve $x = y^2$. To graph this curve, we can start by calculating the values of x for some values of y:

$$\begin{aligned} y=-3 &\longrightarrow x=(-3)^2=9\\ y=-2 &\longrightarrow x=(-2)^2=4\\ y=-1 &\longrightarrow x=(-1)^2=1\\ y=0 &\longrightarrow x=0^2=0\\ y=1 &\longrightarrow x=1^2=1\\ y=2 &\longrightarrow x=2^2=4\\ y=3 &\longrightarrow x=3^2=9 \end{aligned}$$

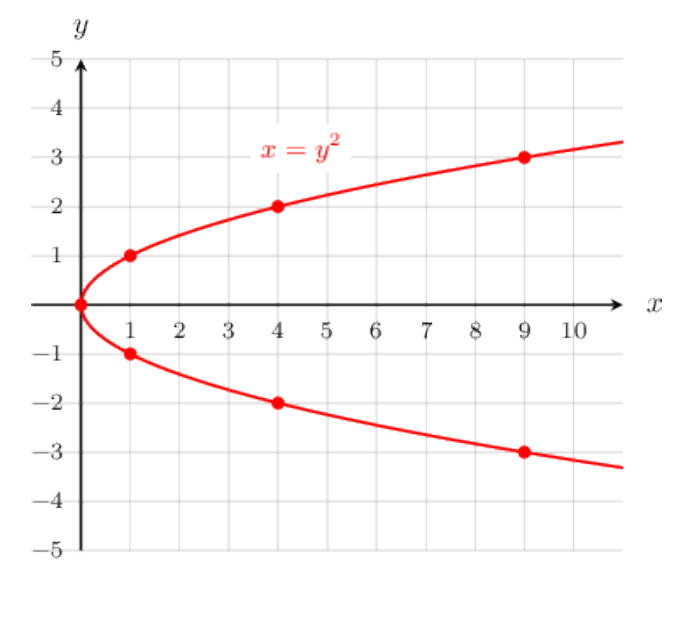
We can summarize these results in the following table:

y	-3	-2	-1	0	1	2	3
x	9	4	1	0	1	4	9

So, we have the following points:

$$(9,-3),\quad (4,-2),\quad (1,-1),\quad (0,0),\quad (1,1),\quad (4,2),\quad (9,3).$$

If we plot these points and connect them with a smooth curve, we get the following graph:



We say that the equation of the curve is defined **implicitly**, because instead of the usual $y = f(x)$, we have $x = f(y)$.

Watch out! Here, we are viewing $x = y^2$ not as a function but rather as an equation. A point belongs to the curve if and only if its coordinates satisfy the equation. The curve does not represent a function since it fails the vertical line test.

Example: Plotting X as a Function of Y Using a Table of Values

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Plot the graph of $x = \frac{1}{2}y - \sqrt{y}$ by completing the following table:

y	0	1	4	16
x				

EXPLANATION

To graph this curve, we can start by calculating the value of x for various values of y, as follows:

$$\begin{aligned} y=0 &\longrightarrow x=\frac{1}{2}\cdot 0-\sqrt{0}=0\\ y=1 &\longrightarrow x=\frac{1}{2}\cdot 1-\sqrt{1}=-\frac{1}{2}\\ y=4 &\longrightarrow x=\frac{1}{2}\cdot 4-\sqrt{4}=0\\ y=16 &\longrightarrow x=\frac{1}{2}\cdot 16-\sqrt{16}=4 \end{aligned}$$

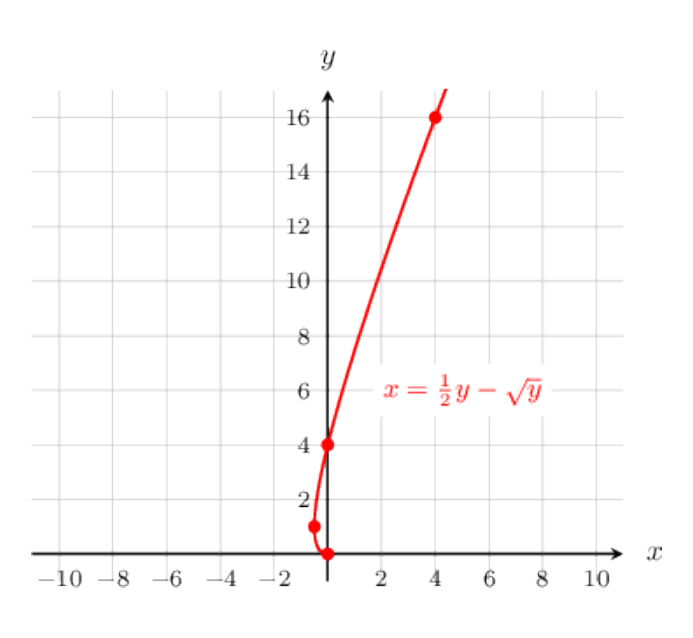
We can summarize these results in the following table:

y	0	1	4	16
x	0	$-\frac{1}{2}$	0	4

So, we have the following points:

$$(0,0),\quad \left(-\frac{1}{2},1\right),\quad (0,4),\quad (4,16).$$

If we plot these points and connect them with a smooth curve, we get the following graph:



KEY PREREQUISITES

Graphing Linear Equations
The Square Root of a Perfect Square

Question 1

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Determine which of the following is the graph of $x = y - 2\sqrt{y}$ by completing the following table:

y	0	1	4	9
x				

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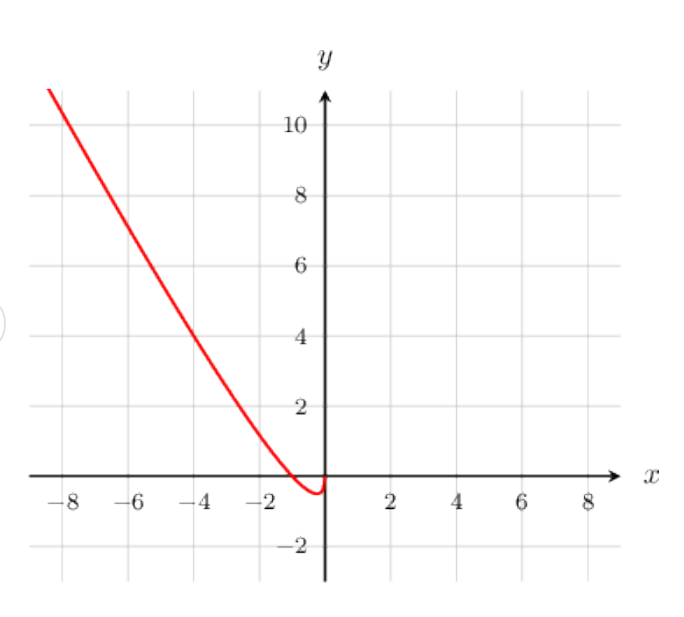

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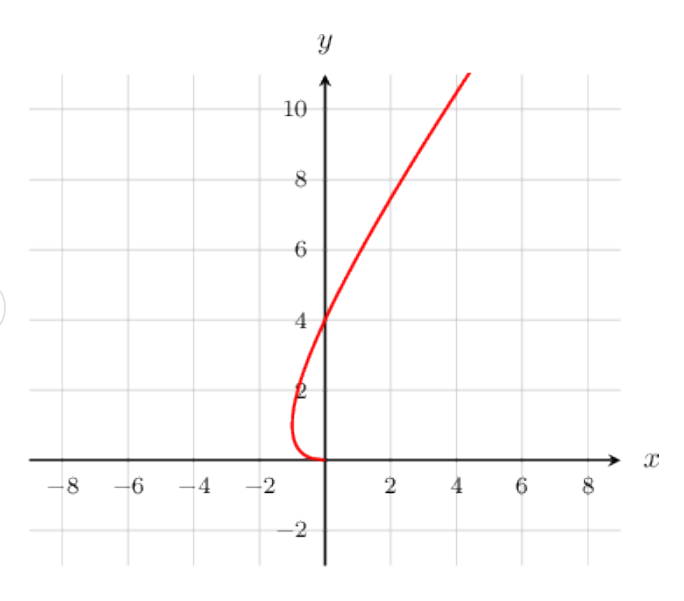

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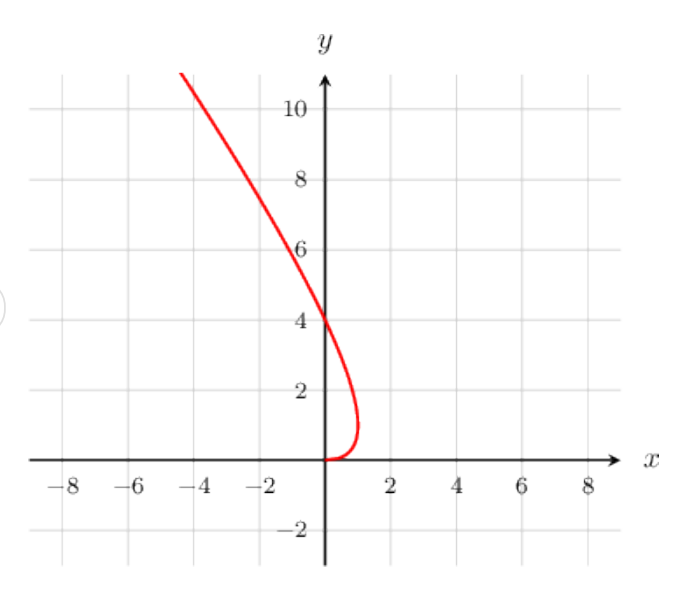

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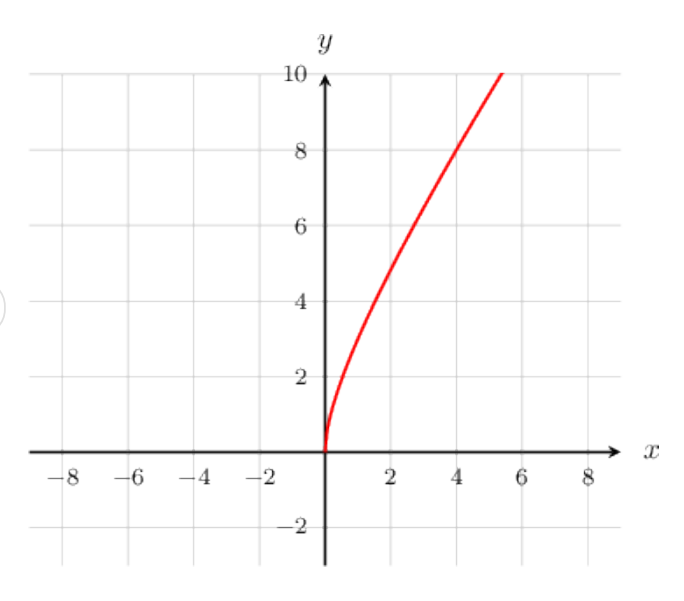

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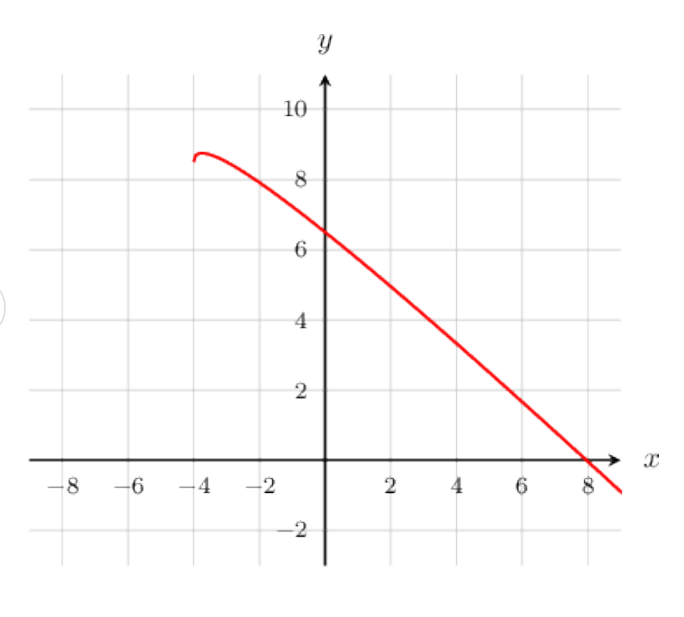

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Question 2

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Determine which of the following is the graph of $x = \frac{5y+10}{y+5}$ by completing the following table:

y	0	1	4	10
x				

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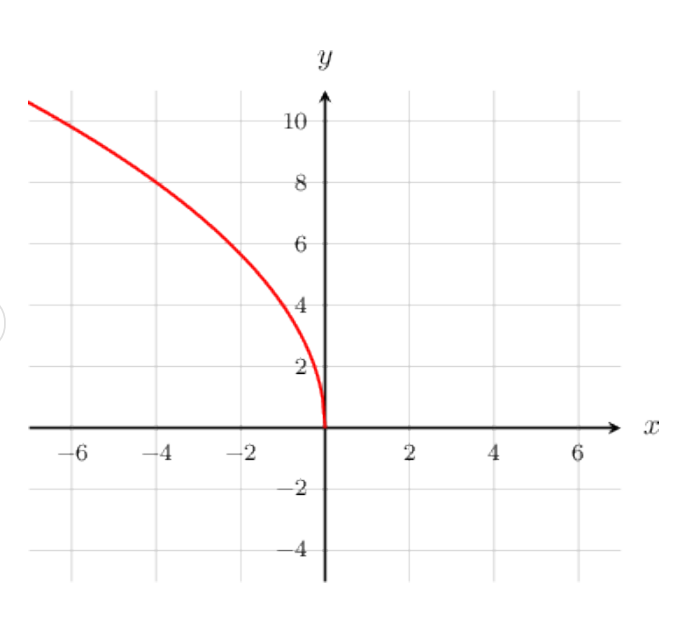

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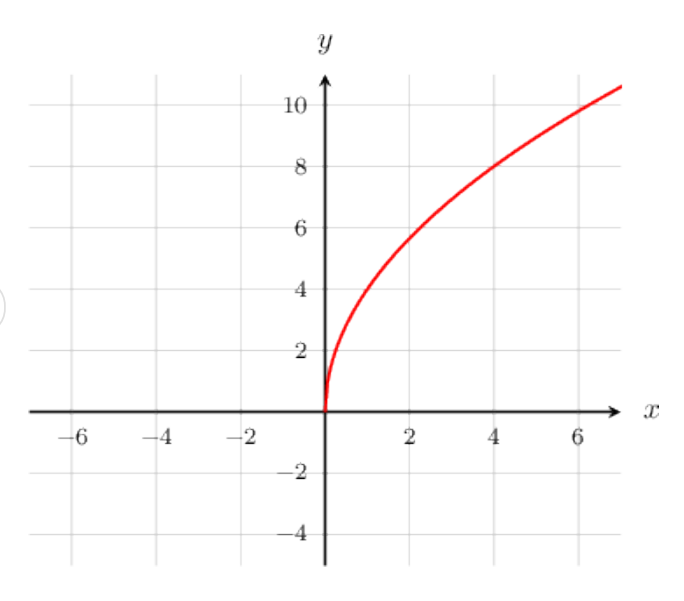

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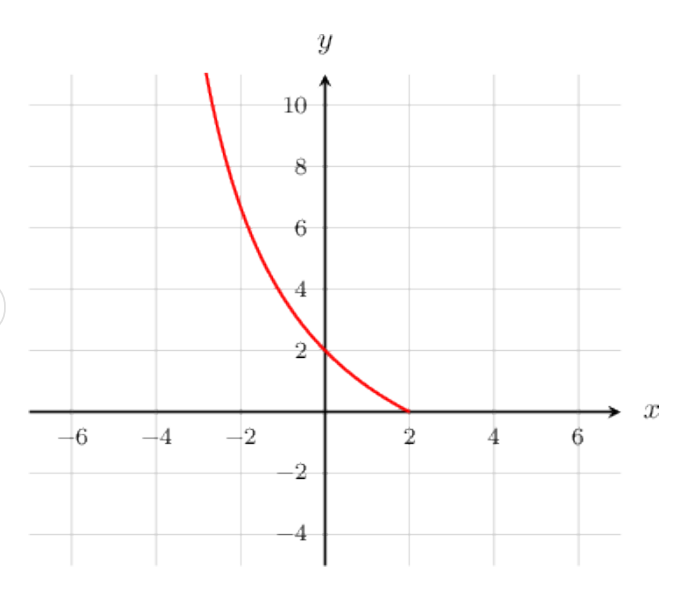

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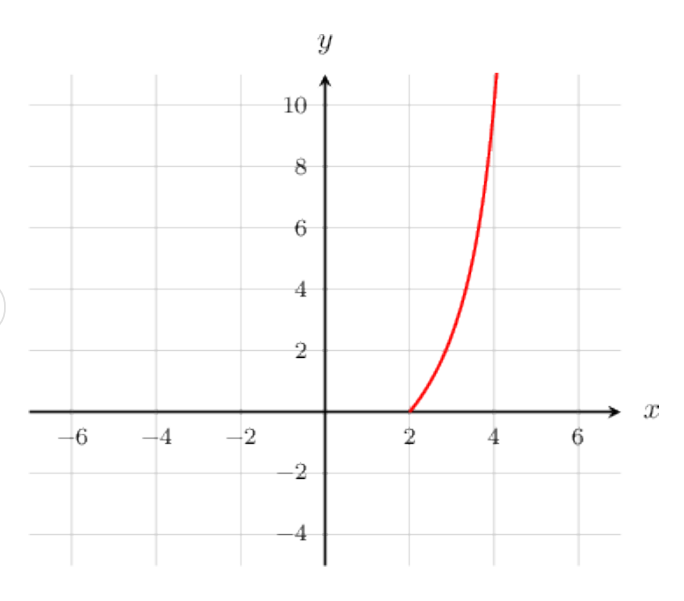

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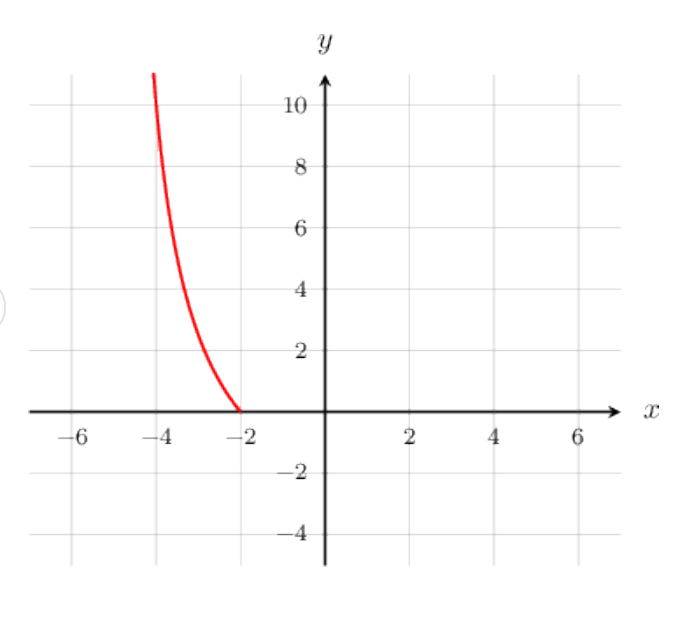

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Plotting X as a Function of Y Using a Reflection

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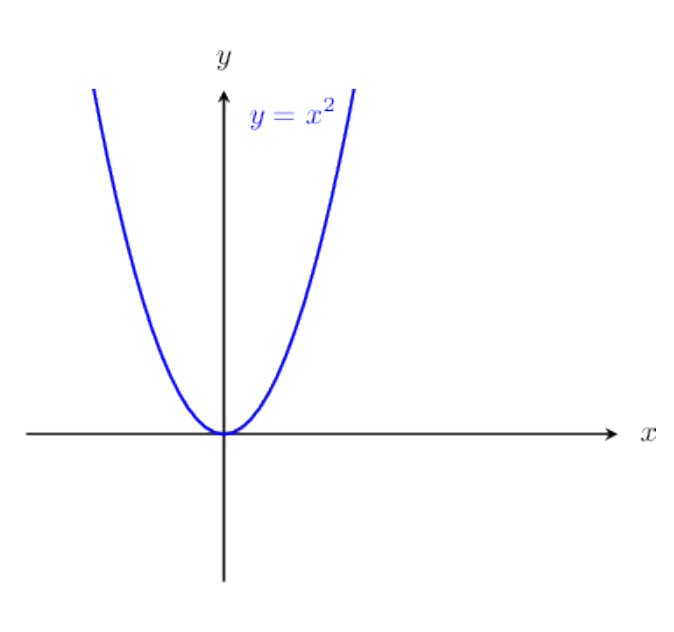
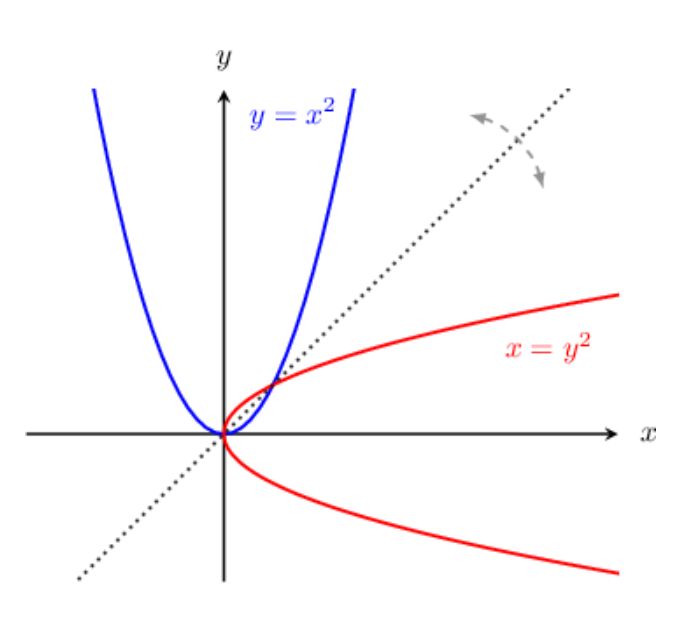
Another way to plot $x = f(y)$ is to first plot the corresponding function $y = f(x)$ and then reflect it over the line $y = x$.

To illustrate, let's use this technique to plot the curve $x = y^2$. To get the corresponding function, we swap x and y and get $y = x^2$. This curve is familiar and much easier to graph:



Now, to get the curve $x = y^2$, we just have to reflect the curve $y = x^2$ over the line $y = x$, as follows:



You may notice that this process is very similar to plotting the inverse of a function. In fact, if $y = f(x)$ is a one-to-one function, then the corresponding function $x = f(y)$ will always give the inverse function.

Example: Plotting X as a Function of Y Given the Graph of the Corresponding Function

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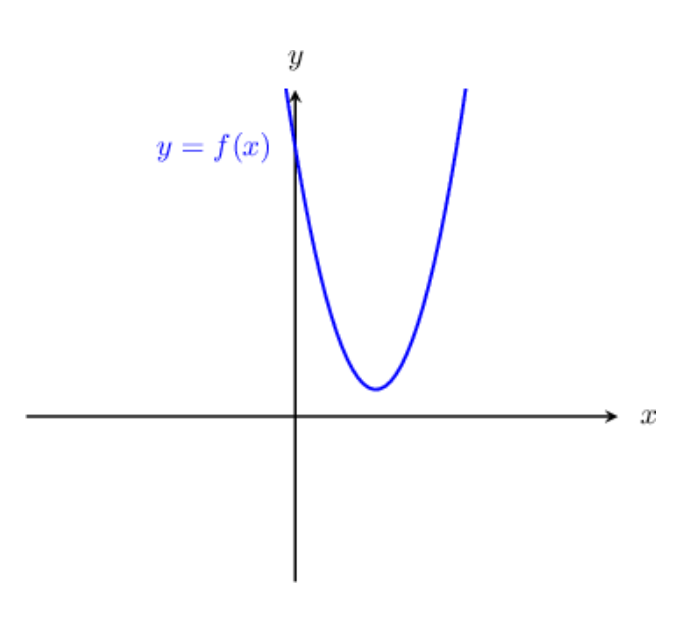
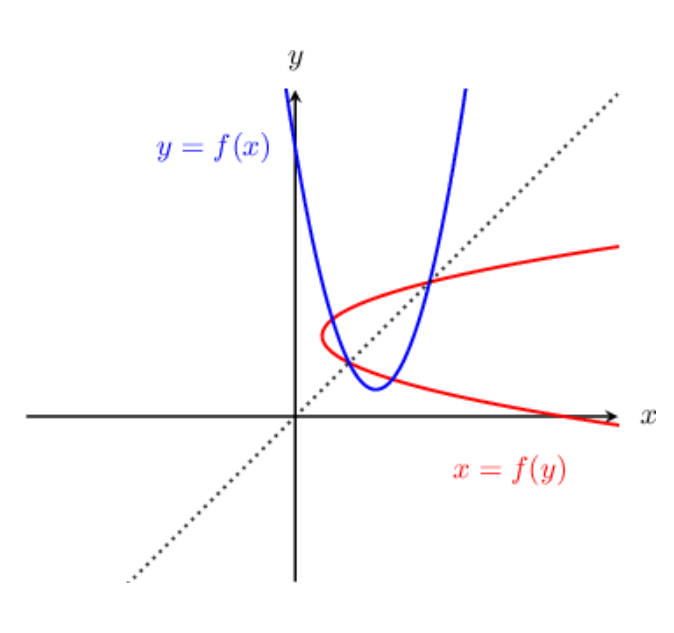
The graph of $y = f(x)$ is shown below. Sketch the graph of $x = f(y)$.



EXPLANATION

To plot the graph of $x = f(y)$, we take the graph of $y = f(x)$ and reflect it over the line $y = x$, as follows:



KEY PREREQUISITE

Graphs of Inverse Functions

Question 3

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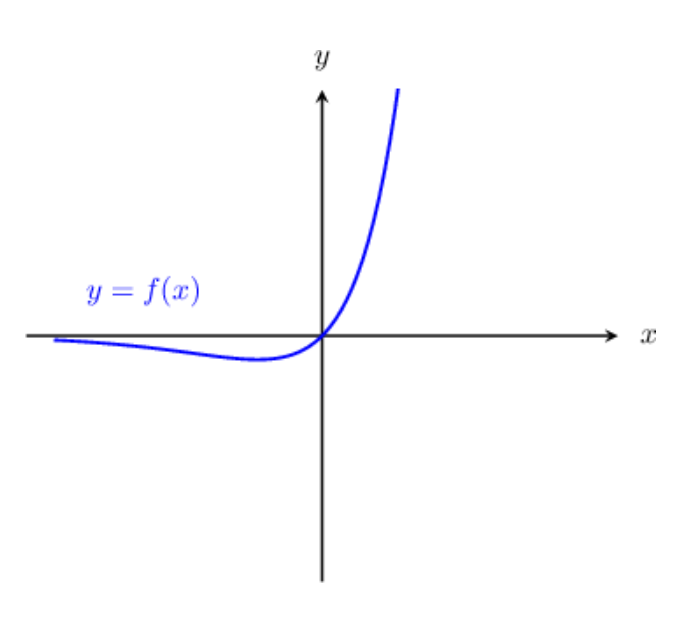

The graph of $y = f(x)$ is shown above. Sketch the graph of $x = f(y)$.

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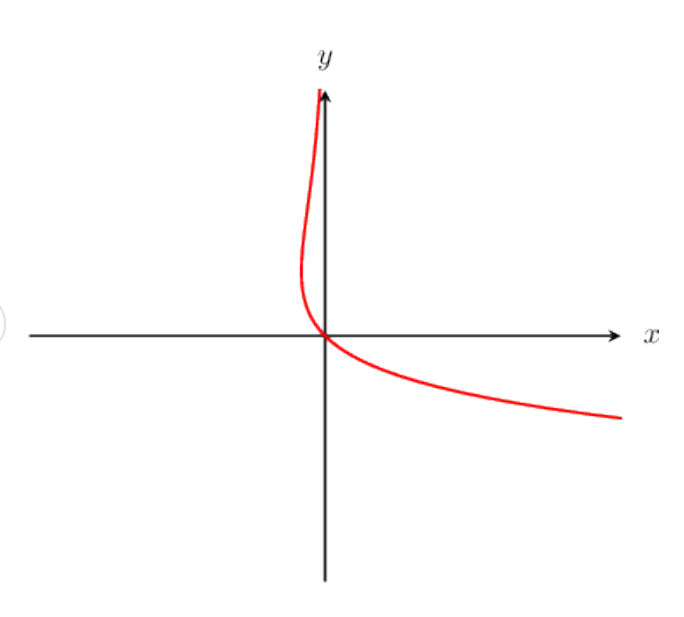

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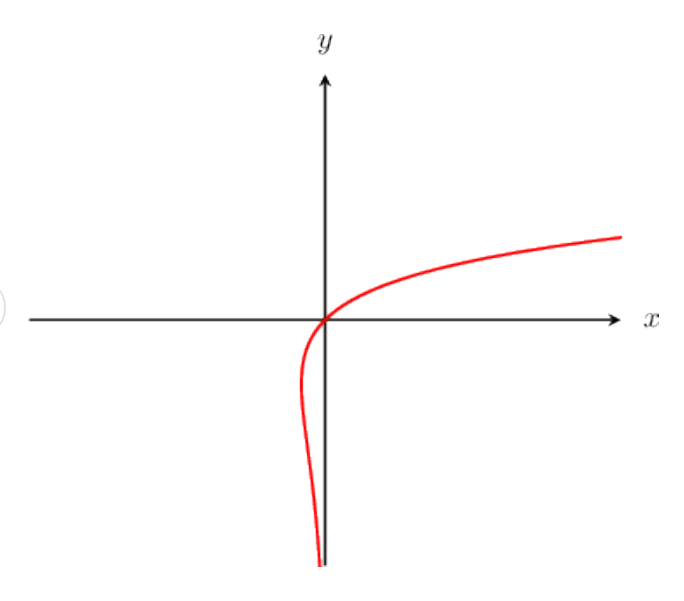

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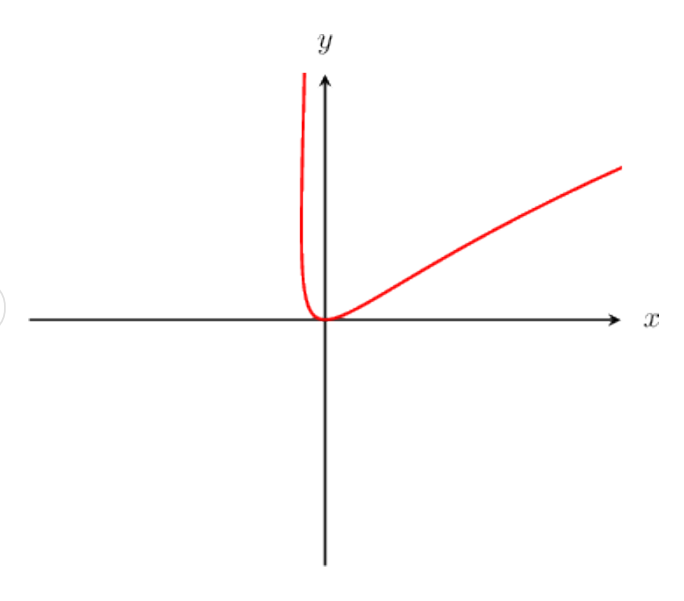

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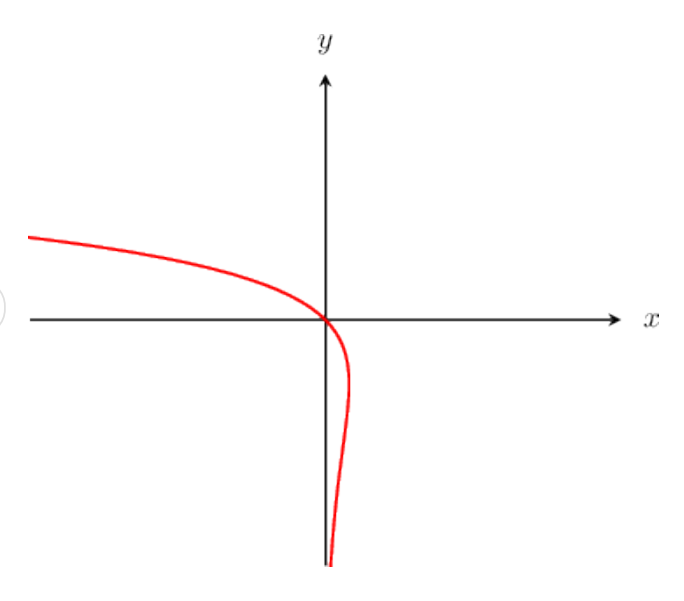

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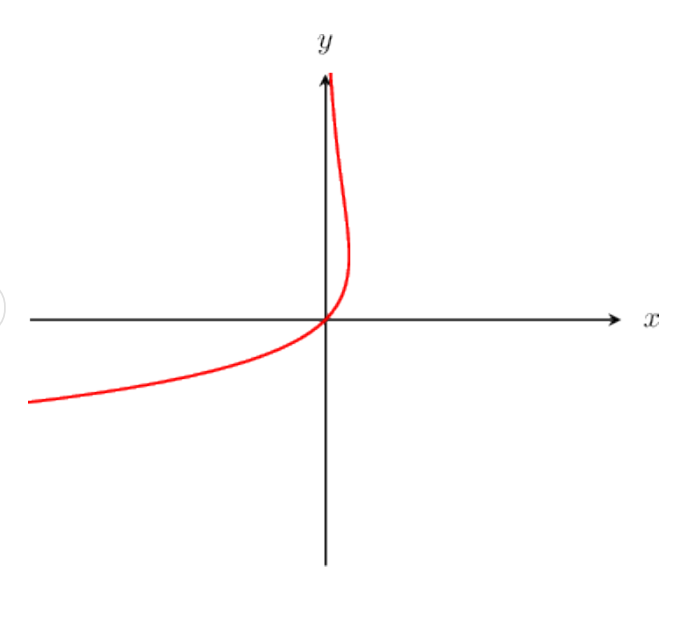

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Question 4

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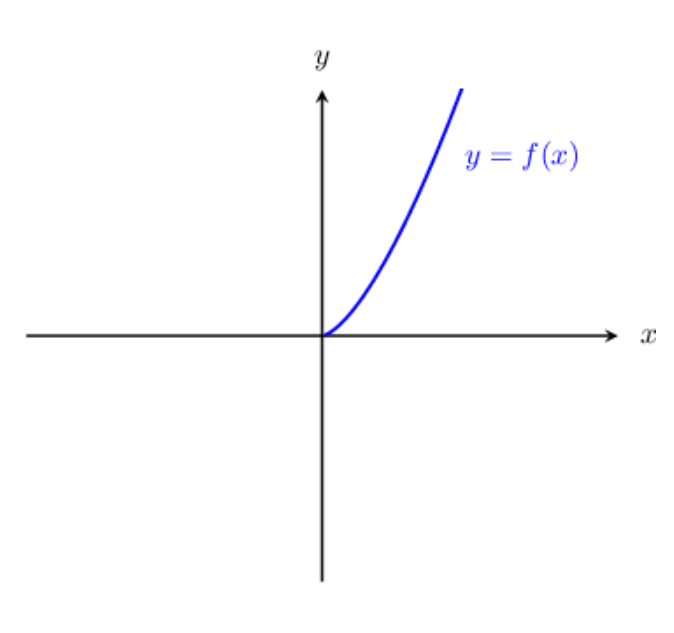

The graph of $y = f(x)$ is shown above. Sketch the graph of $x = f(y)$.

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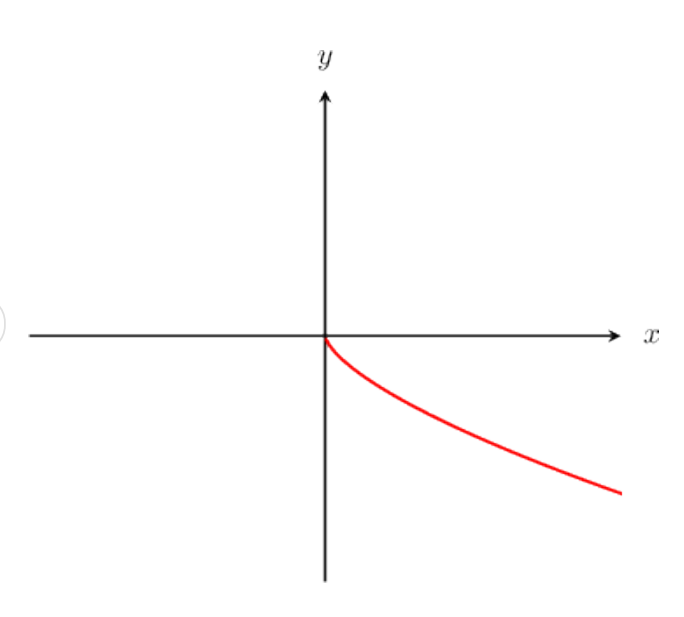

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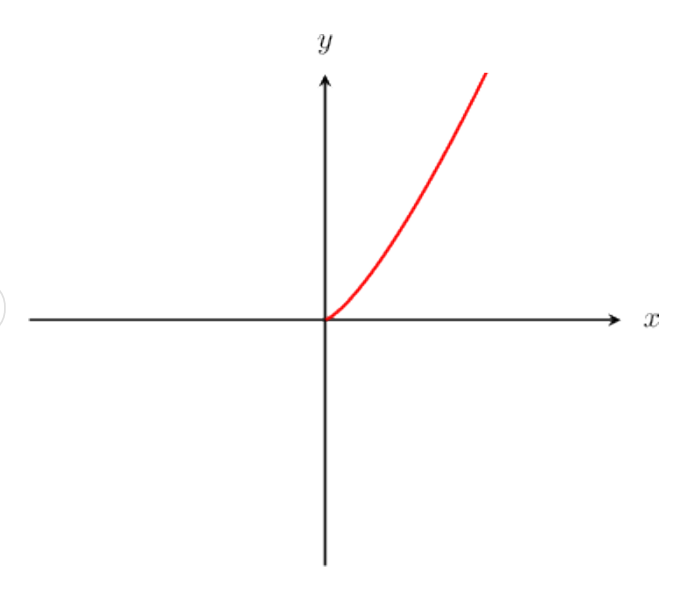

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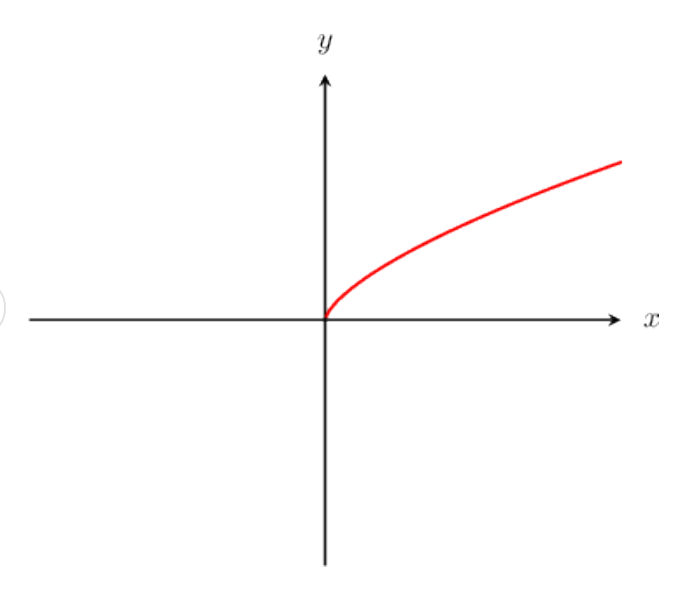

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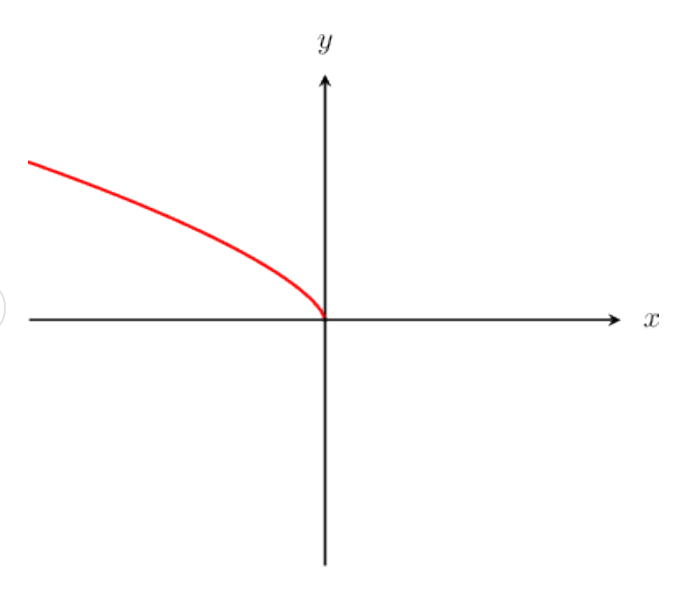

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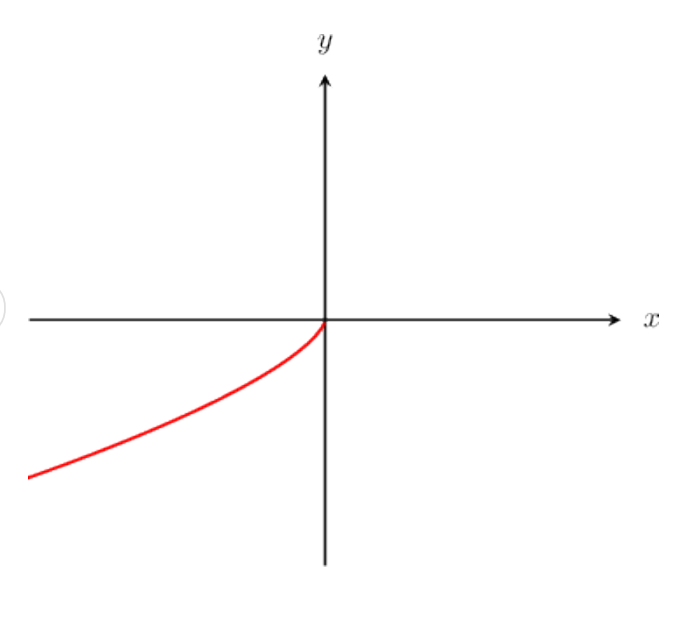

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Example: Plotting a Quadratic Function Where X Is a Function of Y

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Plot the graph of $x = -2y^2 - 3y$.

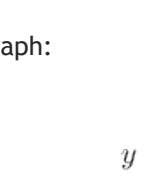
EXPLANATION

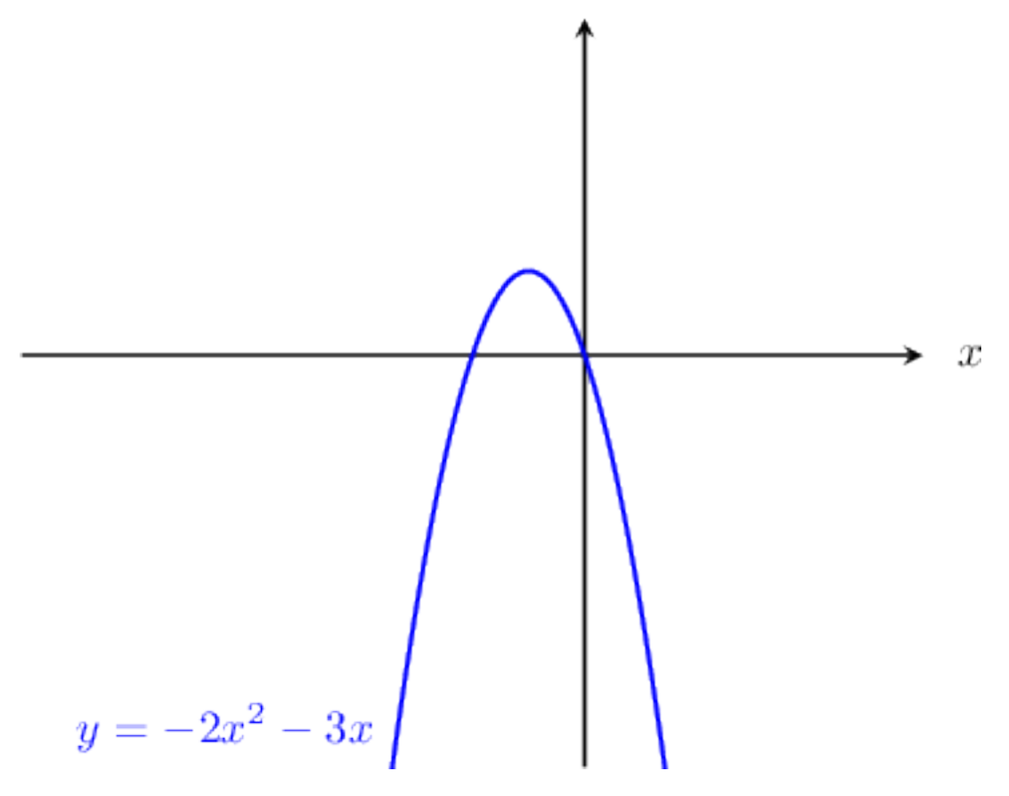
To plot $x = -2y^2 - 3y$, we will first plot the function $y = -2x^2 - 3x$ and then reflect it over the line $y = x$.

The curve $y = -2x^2 - 3x$ is a parabola. It has the following factored form:

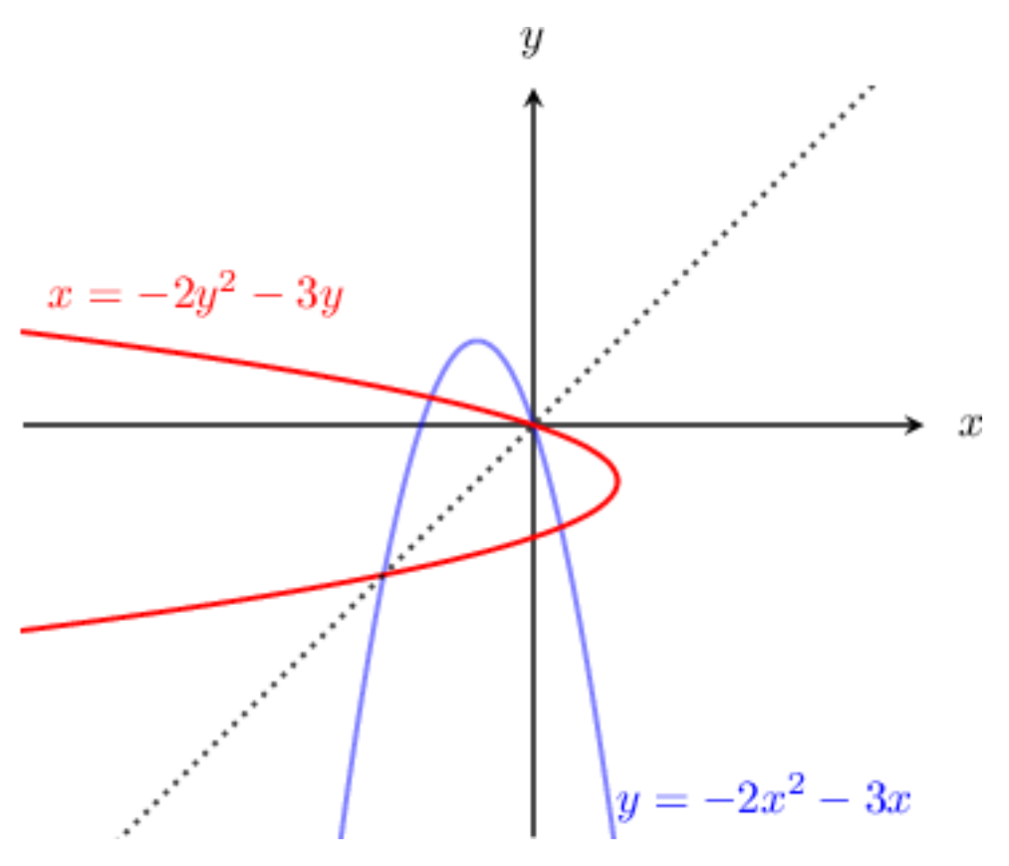
$$y = -x(2x + 3)$$

From the factored form above, we see that we have a negative parabola with roots at $x = 0$ and $x = -\frac{3}{2}$. Therefore, we get the following graph:





Now, to get the curve $x = -2y^2 - 3y$, we just have to reflect the curve $y = -2x^2 - 3x$ over the line $y = x$, as follows:



KEY PREREQUISITE

Roots of Quadratic Functions

Question 5

?

Plot the graph of $x = -y^2 - 3y$.

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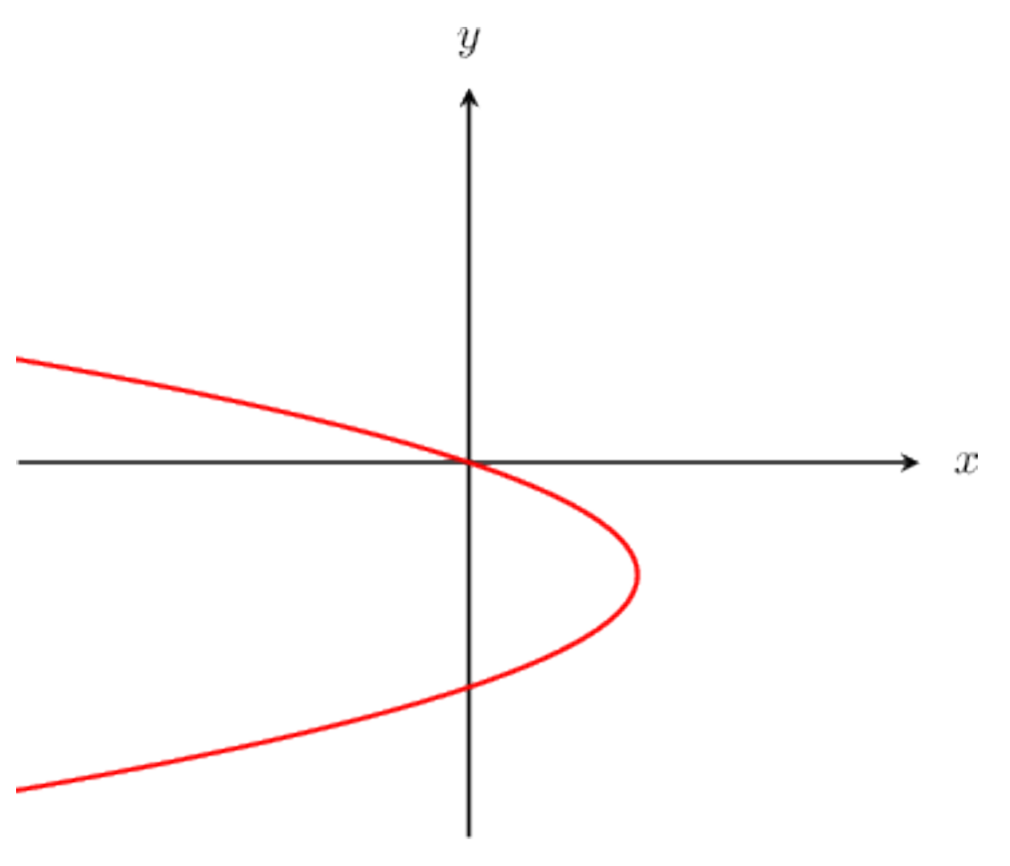

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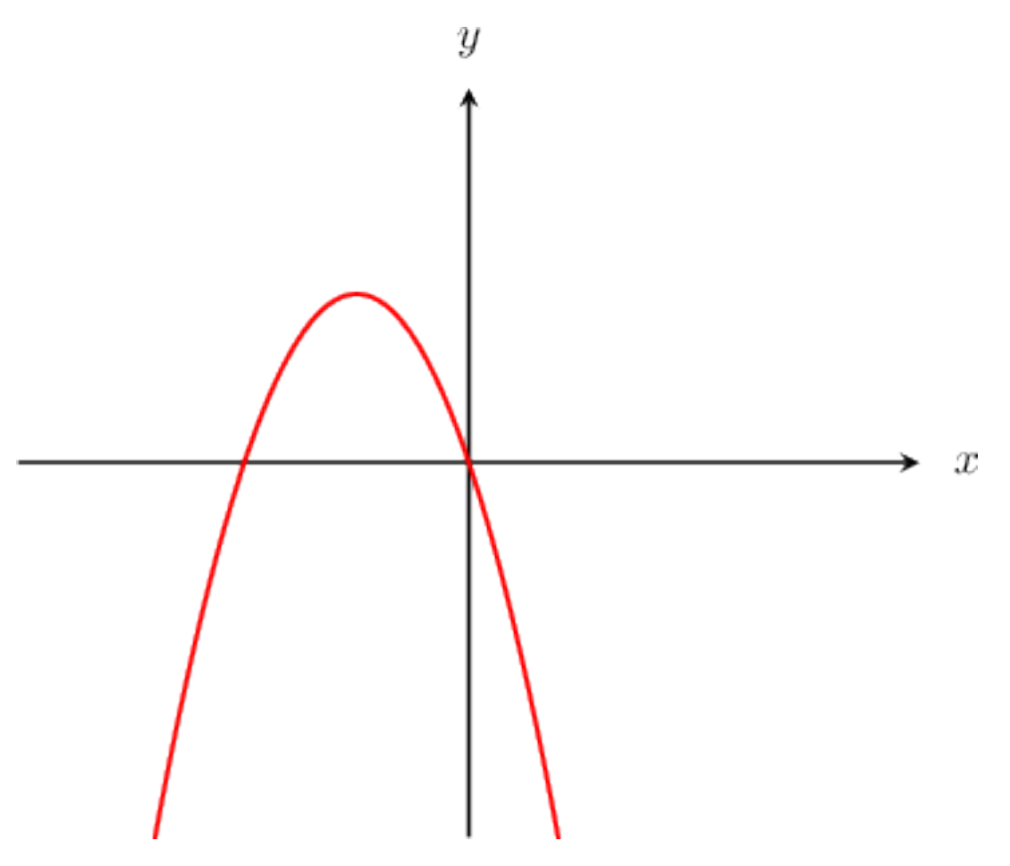

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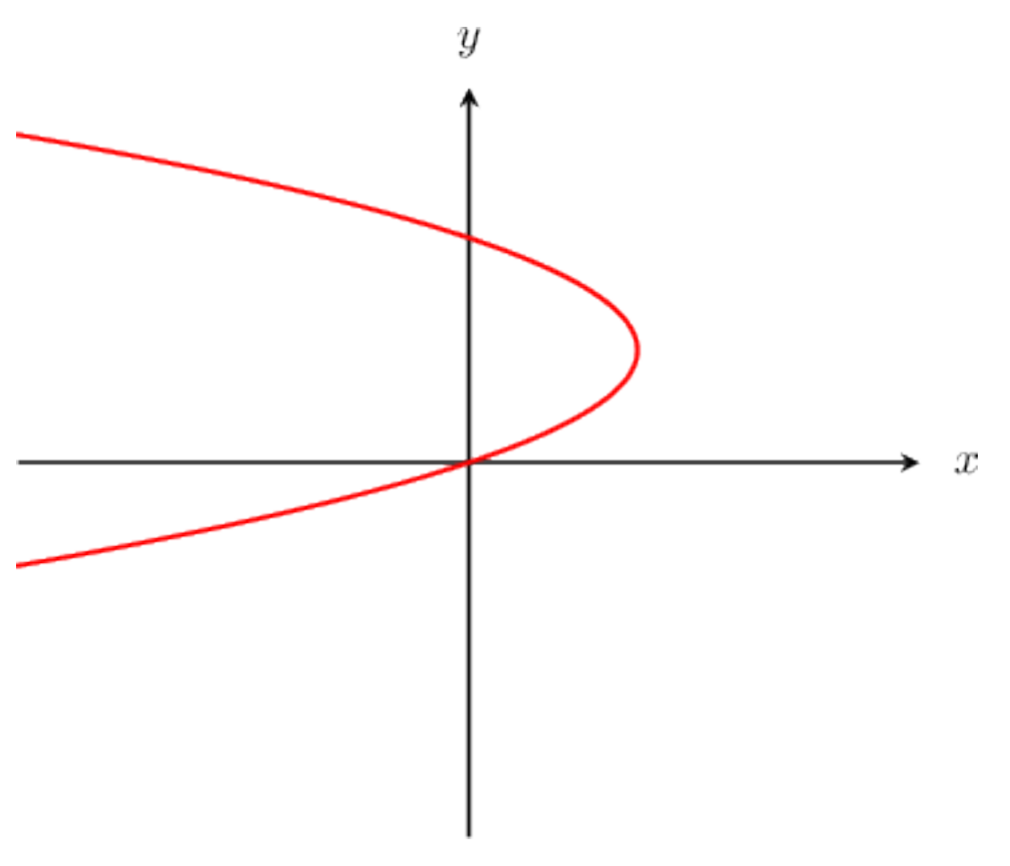

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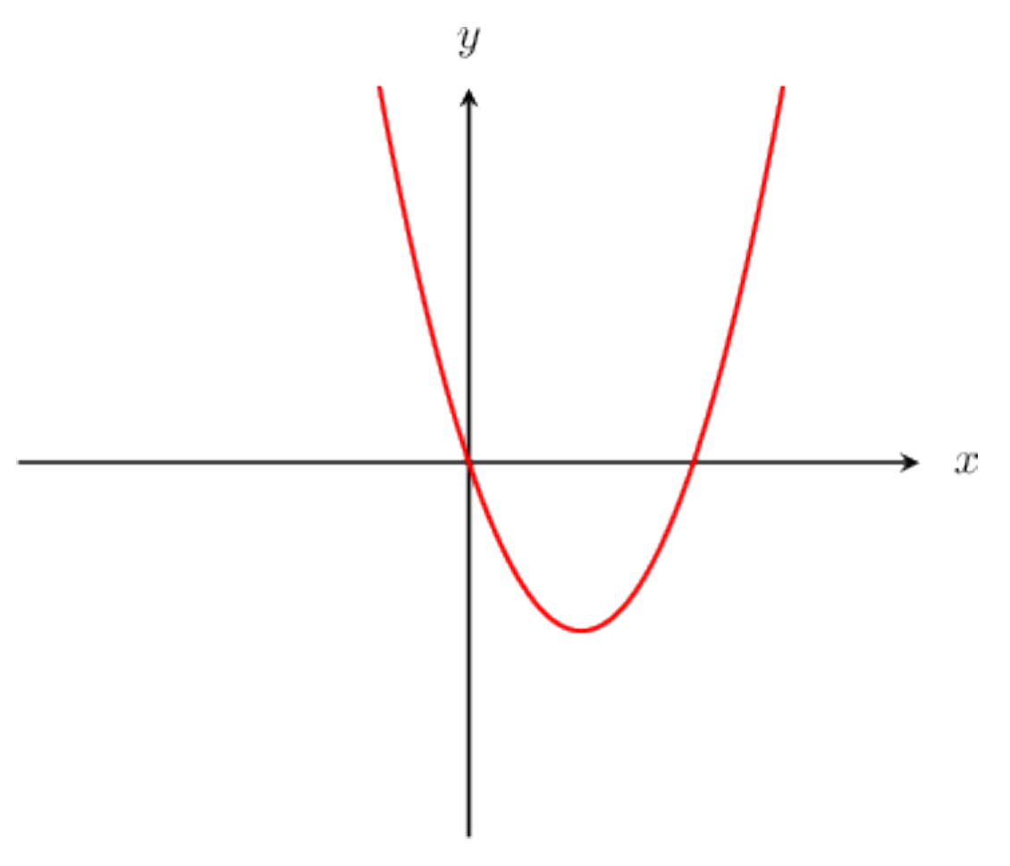

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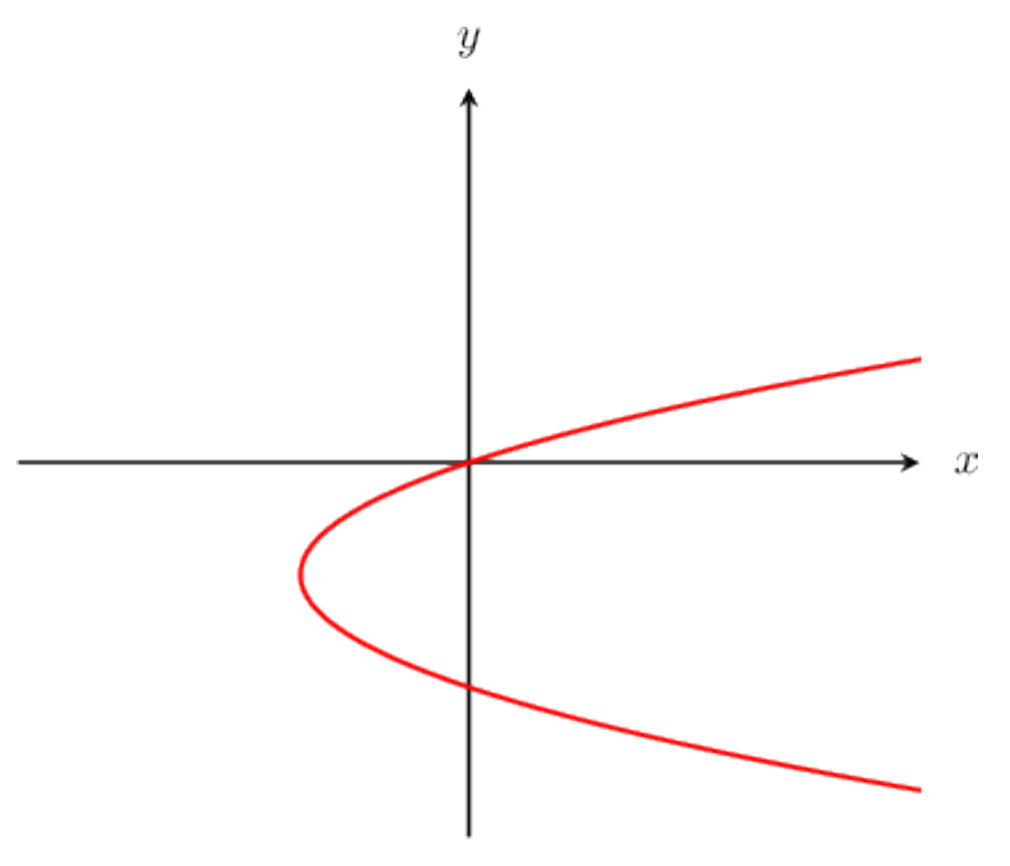

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Question 6

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Plot the graph of $x = y^2 - 2y$.

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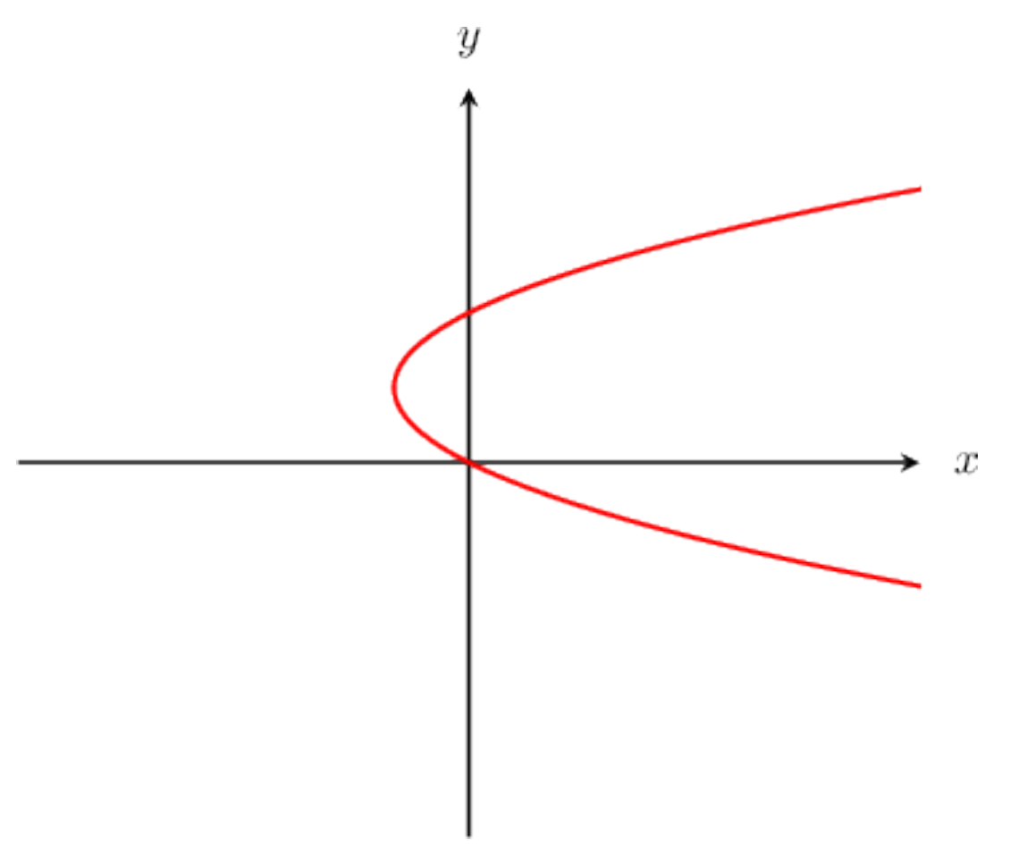

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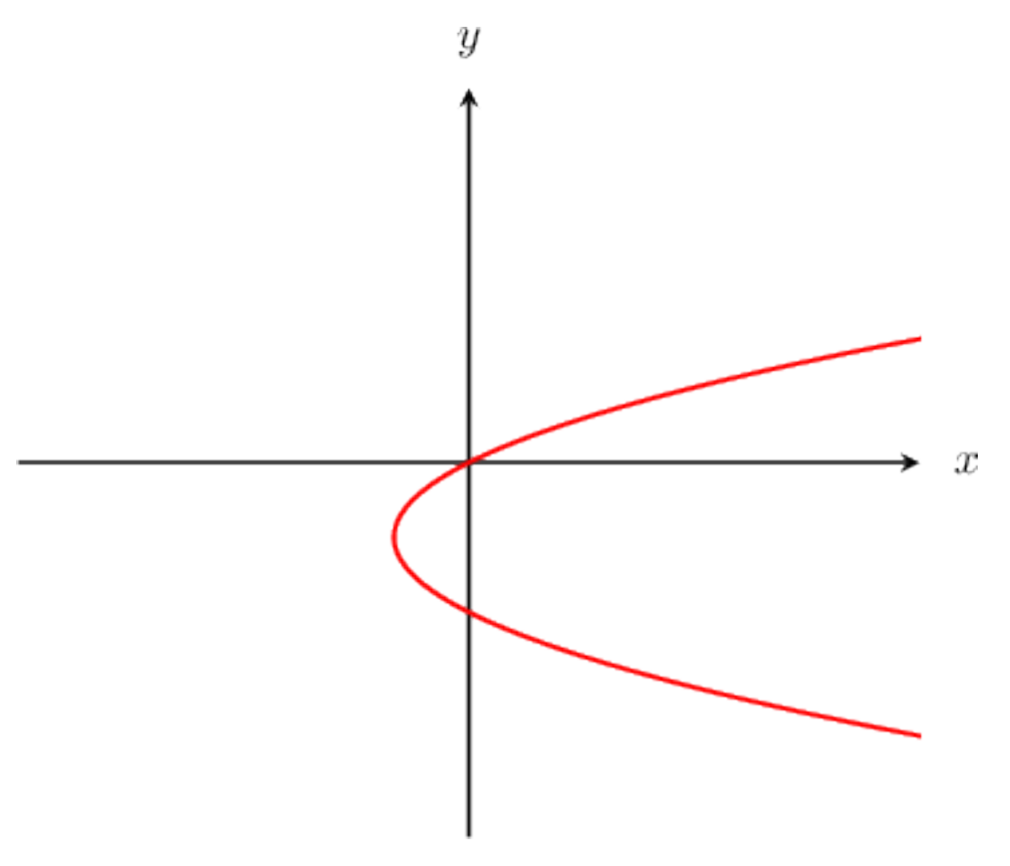

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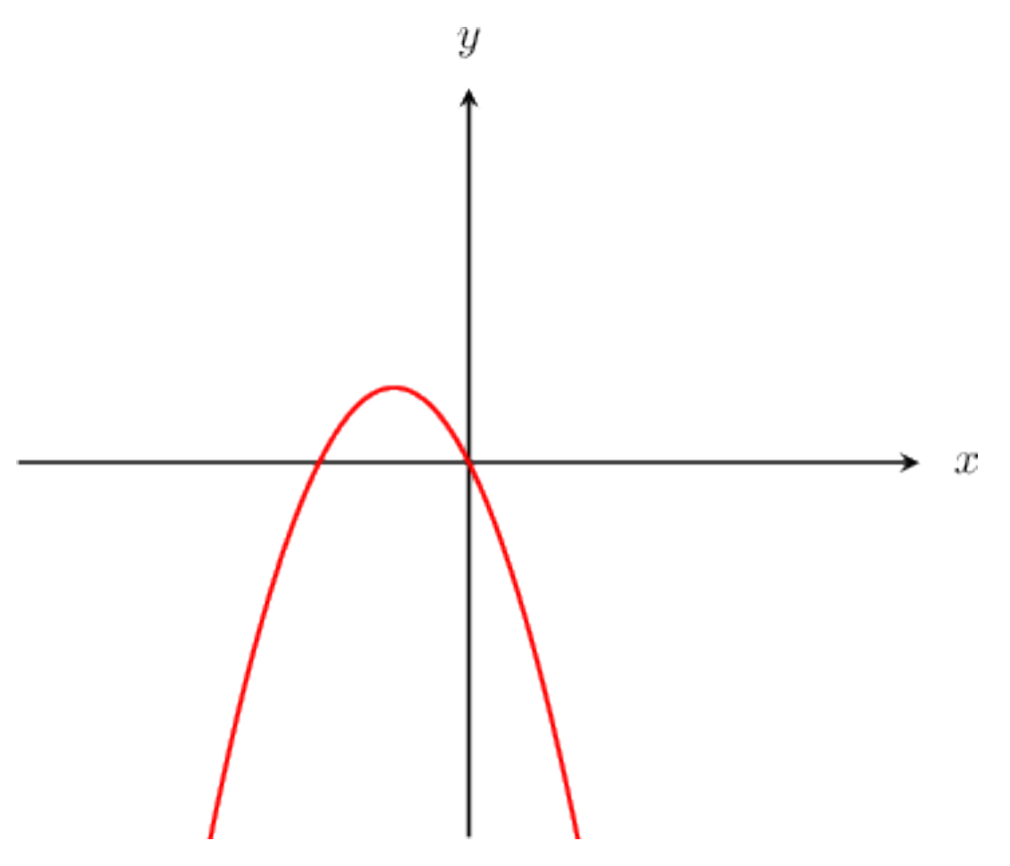

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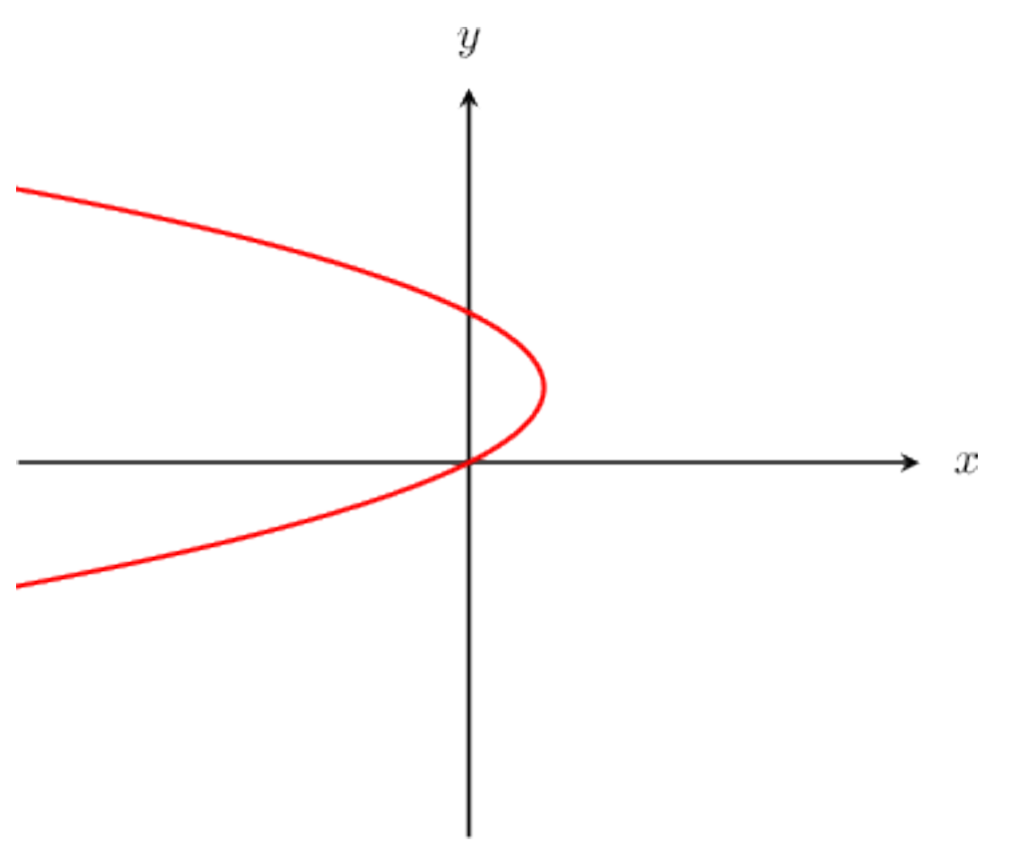

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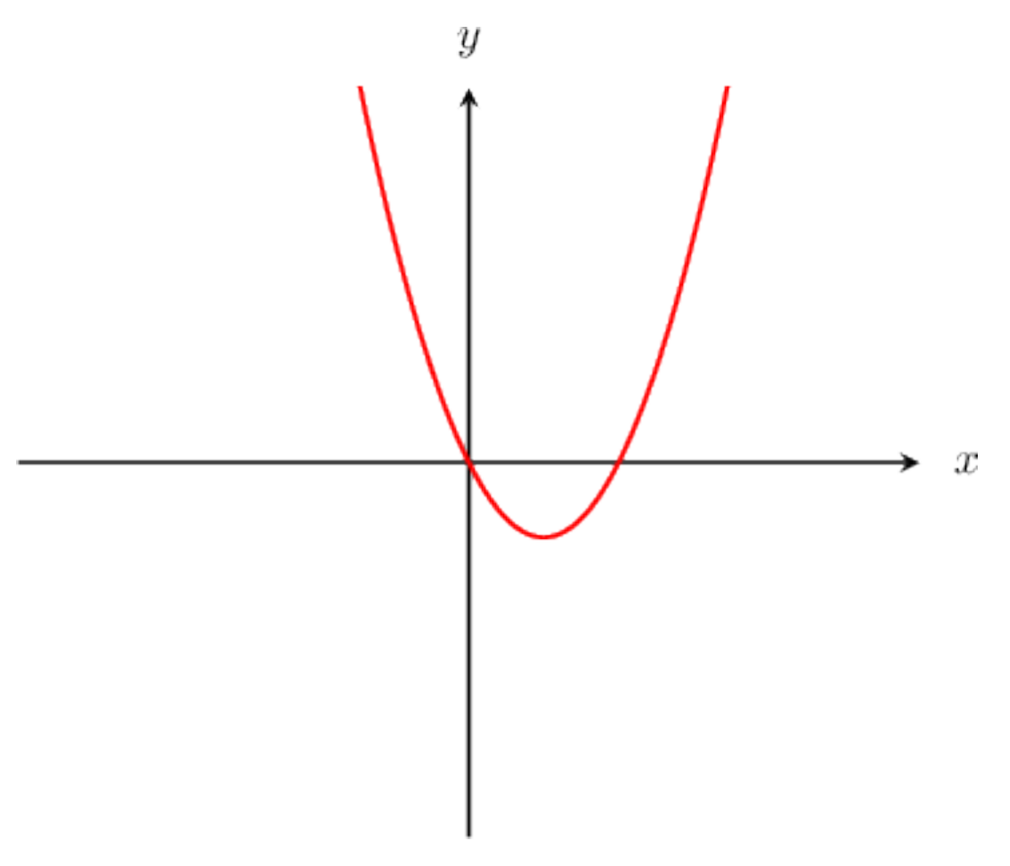

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